

Logarithmic lower bounds for the block complexity of algebraic numbers

Abstract

Let $b \geq 2$ be an integer, let ξ be a real algebraic irrational number, and let $p_{\xi,b}(L)$ count the distinct blocks of length L in the fractional base- b expansion of ξ . We prove that

$$\limsup_{L \rightarrow \infty} \frac{p_{\xi,b}(L)(\log \log L)^\gamma}{L(\log L)^{1/3}} = +\infty \quad \text{for every } \gamma > 1.$$

In particular, $p_{\xi,b}(L)/(L(\log L)^\eta)$ is unbounded for every real $\eta < 1/3$. The proof combines rational approximations obtained from repeated blocks with the twisted-height interval theorem of Evertse and Ferretti. A selection of the approximations has uniformly bounded intersection with every proper linear subspace. This bound follows from a divisibility condition and an algebraic norm estimate. For composite bases, the approximating vectors need not be primitive; a direct use of the external parameter in the twisted height avoids normalisation by their common divisors.

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1 Introduction

Let $b \geq 2$ be an integer and write the fractional part of a real irrational number ξ as

$$\{\xi\} = \sum_{i \geq 1} d_i b^{-i}, \quad d_i \in \{0, \dots, b-1\}.$$

Its block complexity is

$$p_{\xi,b}(L) = \#\{d_{j+1} \cdots d_{j+L} : j \geq 0\}, \quad L \geq 1.$$

Normality to base b implies $p_{\xi,b}(L) = b^L$ for every L . Much weaker lower bounds are already of interest when ξ is algebraic.

Adamczewski and Bugeaud [1, Theorem 1] proved that, for every algebraic irrational ξ and every integer $b \geq 2$,

$$\lim_{L \rightarrow \infty} \frac{p_{\xi,b}(L)}{L} = +\infty.$$

Bugeaud and Evertse [2, Theorem 2.1] established the logarithmic refinement

$$\limsup_{L \rightarrow \infty} \frac{p_{\xi,b}(L)}{L(\log L)^\eta} = +\infty \quad (\eta < 1/11).$$

The result proved here is the following.

Theorem 1.1. *For every fixed integer $b \geq 2$, every real algebraic irrational number ξ , and every real $\eta < 1/3$, one has*

$$\limsup_{L \rightarrow \infty} \frac{p_{\xi,b}(L)}{L(\log L)^\eta} = +\infty. \quad (1.1)$$

Equivalently, for every $C > 0$ there are arbitrarily large integers L such that $p_{\xi,b}(L) > CL(\log L)^\eta$.

Retaining a secondary logarithmic factor gives the following refinement.

Theorem 1.2. *For every fixed integer $b \geq 2$, every real algebraic irrational number ξ , and every real $\gamma > 1$, one has*

$$\limsup_{L \rightarrow \infty} \frac{p_{\xi,b}(L)(\log \log L)^\gamma}{L(\log L)^{1/3}} = +\infty. \quad (1.2)$$

The logarithms in (1.2) are considered only for sufficiently large L . This result implies Theorem 1.1, since $(\log L)^{1/3-\eta}/(\log \log L)^\gamma \rightarrow \infty$ when $\eta < 1/3$. Both conclusions are limsup statements. The unweighted endpoint $\limsup p_{\xi,b}(L)/(L(\log L)^{1/3}) = +\infty$ and analogous lower bounds at every sufficiently large length remain open here, as do positive entropy, disjointivity, and normality. Bugeaud and Evertse also note a secondary logarithmic refinement at their exponent $1/11$ in [2, equation (2.4)].

We follow the repetition construction underlying [2, §6]. An eventual bound $p_{\xi,b}(L) \ll L(\log L)^u$ supplies approximations

$$|(b^t - b^r)\xi - a| \leq b^{-v}, \quad v \gg \ell(\log \ell)^{-u}, \quad 0 \leq r < t \leq \ell.$$

Choosing a shortest preperiod gives $b \nmid a$ whenever $r > 0$. We then select a sequence for which the period lengths $s = t - r$ are distinct and the exponents t more than double. The selected vectors (b^t, b^r, a) have uniformly bounded intersection with every proper subspace. The bound depends only on b and ξ , and not on the approximation exponent; see Proposition 4.1.

This intersection bound replaces the further application of a two-dimensional approximation theorem used in [2, §7]. We combine it with the interval form of the quantitative Subspace Theorem in [3, Theorem 2.3]. The uniform intersection bound controls the exceptional subspace independently of the changing weights. The remaining count has the following three factors, shown by their orders of magnitude:

$$\underbrace{\varepsilon^{-1}}_{\text{ratio intervals}} \underbrace{\varepsilon^{-2} \log(2/\varepsilon)}_{\text{EF intervals}} \underbrace{\log(2/\varepsilon)}_{\text{points per interval}} = \varepsilon^{-3}(\log(2/\varepsilon))^2.$$

The exceptional subspaces contribute only $O_{b,\xi}(\varepsilon^{-1})$ additional points. Thus substituting a sharper quantitative theorem alone is not the whole argument: the uniform subspace bound and the selection of period lengths are also used. Together with the growth of the selected prefix lengths, this gives the range $u < 1/3$ and the refinement in Theorem 1.2. Proposition 8.1 explains the cost of the ratio partition within the fixed-weight estimates used here.

All arguments below apply to each fixed integer base. In composite bases, $b \nmid a$ does not imply $\gcd(a, b) = 1$. Consequently, the size b^t cannot simply be identified with the projective height of the vector. Section 6 gives the local inequalities and their conversion to twisted height with an external parameter. Appendix A records the weaker exponents obtained from covering versions of the Subspace Theorem.

Throughout, logarithms are natural unless a base is specified. Subtracting the integer part, we assume from now on that $0 < \xi < 1$, and put $d = [\mathbb{Q}(\xi) : \mathbb{Q}]$. For $x \in \mathbb{Z}^3 \setminus \{0\}$, its absolute projective height is

$$H(x) = \frac{\max_i |x_i|}{\gcd(x_1, x_2, x_3)}.$$

Implied constants may depend on parameters explicitly fixed in the argument. They are independent of the varying approximation exponent, the chosen interval of ratios, the selected sequence length, and the exceptional subspace.

2 Repeated blocks and rational approximations

Fix $u > 0$ and $\gamma \geq 0$, and define, for real $x > e$,

$$F(x) = \frac{(\log x)^u}{(\log \log x)^\gamma}. \quad (2.1)$$

The case $\gamma = 0$ gives the original logarithmic bound.

Lemma 2.1. *The function F is eventually increasing, tends to infinity, and satisfies $F(x) = o(x)$. For every fixed $D > 0$,*

$$\frac{F(\lceil Dx F(x) \rceil)}{F(x)} \longrightarrow 1 \quad (x \rightarrow \infty). \quad (2.2)$$

Proof. For sufficiently large x ,

$$\frac{d}{dx} \log F(x) = \frac{u - \gamma / \log \log x}{x \log x} > 0.$$

Also $\log F(x) = u \log \log x - \gamma \log \log \log x$ tends to infinity and is $o(\log x)$, which proves the asserted growth properties. Writing $x^* = \lceil Dx F(x) \rceil$, we have $\log x^* = \log x + O_{D,u,\gamma}(\log \log x)$. Thus both $\log x^* / \log x$ and $\log \log x^* / \log \log x$ tend to 1. Substituting these two ratios in (2.1) proves (2.2). $\square$

Suppose that, for some fixed $C \geq 1$ and sufficiently large k_0 ,

$$p_{\xi,b}(k) \leq CkF(k) \quad (k \geq k_0). \quad (2.3)$$

Increase k_0 so that F is increasing, $F \geq 1$, and $\log \log x \geq 1$ for $x \geq k_0$. We use juxtaposition for concatenation of finite words; empty words are allowed except where a positive length is specified. Let

$$A_j = \sum_{i=1}^j d_i b^{j-i} \quad (j \geq 1), \quad A_0 = 0.$$

Lemma 2.2. *For every sufficiently large integer ℓ , there is a decomposition*

$$d_1 \cdots d_\ell = UVWVX, \quad v := |V| \geq a_0 \ell / F(\ell), \quad a_0 = (24C)^{-1}. \quad (2.4)$$

Proof. Put $k = \lfloor \ell / (4CF(\ell)) \rfloor$. For large ℓ , one has $k \geq k_0$ and

$$p_{\xi,b}(k) \leq CkF(k) \leq \ell/4,$$

using the monotonicity of F . There are $\ell - k + 1 > \ell/4$ starting positions for length- k blocks in the prefix. Two of these blocks coincide. Let D be the distance between their starting positions.

If $D \geq k$, the two occurrences are disjoint. If $D < k$, the substring of length $k + D$ containing them has period D . Its first two consecutive blocks of length

$$v' = D \left\lfloor \frac{k + D}{2D} \right\rfloor$$

are equal, since D divides v' and $2v' \leq k + D$. Moreover $v' \geq k/3$. Indeed, if $D \geq k/3$ then $v' \geq D$; otherwise $v' > (k - D)/2 > k/3$. In either case there is a disjoint repeated block of length at least $k/3$. For large ℓ , the bound $k \geq \ell/(8CF(\ell))$ proves (2.4). $\square$

For each such ℓ , choose a decomposition (2.4) with $|V|$ maximal, and among these choose one with $|U|$ minimal. Any remaining choice is fixed arbitrarily. Write

$$r = |U|, \quad s = |VW|, \quad t = r + s, \quad a = A_t - A_r, \quad q = b^t - b^r.$$

Lemma 2.3. *These integers satisfy*

$$s \geq v, \quad 0 \leq r < t \leq \ell, \quad 0 \leq a \leq q < b^t, \quad |q\xi - a| \leq b^{-v}. \quad (2.5)$$

If $r > 0$, then $b \nmid a$.

Proof. Suppose first that $r > 0$ and $d_r = d_t$. Moving both occurrences of V one position to the left produces two equal blocks of the same length and with the same separation. They remain disjoint, and their first starting position is earlier. This contradicts the minimality of r . Hence $d_r \neq d_t$, and

$$a = A_t - A_r \equiv d_t - d_r \not\equiv 0 \pmod{b}.$$

If P is the base- b value of the word VW , then $A_t = b^s A_r + P$, and therefore

$$\frac{a}{q} = \frac{A_r}{b^r} + \frac{P}{b^r(b^s - 1)}. \quad (2.6)$$

The right side is the value of the infinite word $U(VW)^\infty$. This word and the expansion of ξ have the same first $t + v$ digits. Values with a given prefix of length $t + v$ lie in a closed interval of length $b^{-(t+v)}$. This observation also covers a periodic tail consisting entirely of the digit $b - 1$. Thus

$$|\xi - a/q| \leq b^{-(t+v)}, \quad |q\xi - a| \leq b^{-v}.$$

Finally $0 \leq A_r \leq b^r - 1$ and $0 \leq P \leq b^s - 1$ give

$$0 \leq a = (b^s - 1)A_r + P \leq b^r(b^s - 1) = q < b^t.$$

The remaining inequalities follow from the word lengths. $\square$

3 Selection of the approximations

Attach subscripts ℓ to the quantities constructed in Section 2.

Lemma 3.1. *There is an increasing sequence of admissible prefix lengths ℓ_j such that, on writing $r_j = r_{\ell_j}$ and similarly for s_j, t_j, v_j, a_j ,*

$$s_{j+1} > 2t_j, \quad t_{j+1} > 2t_j, \quad s_{j+1} > s_j. \quad (3.1)$$

For constants $D_1, A > 0$ independent of j , one also has

$$\ell_{j+1} \leq D_1 \ell_j F(\ell_j), \quad \log \ell_j \leq A j \log(2j). \quad (3.2)$$

Proof. Choose ℓ_1 sufficiently large. Given ℓ_j , let ℓ_{j+1} be the least integer $\ell > \ell_j$ for which $s_\ell > 2t_{\ell_j}$. Such an integer exists because $s_\ell \geq a_0 \ell / F(\ell) \rightarrow \infty$. Since $t_j \geq s_j$, this rule implies all of (3.1).

Fix $D > 4/a_0$ and consider $\ell^* = \lceil D \ell_j F(\ell_j) \rceil$. By Lemma 2.1, $F(\ell^*)/F(\ell_j) \rightarrow 1$. Consequently, for all sufficiently large ℓ_j ,

$$s_{\ell^*} \geq a_0 \ell^* / F(\ell^*) \geq \frac{a_0 D}{2} \ell_j > 2t_j.$$

Thus $\ell_{j+1} \leq \ell^*$, proving the first estimate in (3.2) after increasing a fixed constant.

Since $F(x) \leq (\log x)^u$ on the range in question, for $y_j = \log \ell_j$ this estimate implies $y_{j+1} \leq y_j + u \log y_j + O(1)$. Choose A large enough to cover y_1 and to ensure, for every $j \geq 1$,

$$u \log(Aj \log(2j)) + O(1) \leq A \log(2j).$$

This is possible since the left side grows logarithmically in both j and A , whereas the coefficient on the right is linear in A . Induction, using

$$(j+1) \log(2j+2) - j \log(2j) \geq \log(2j),$$

now gives $y_j \leq Aj \log(2j)$. The constants may depend on C, u, γ, k_0 and the initial choice ℓ_1 . $\square$

Since $t_j \geq 2^{j-1} t_1$ and $\ell_j \geq t_j$, the same sequence also satisfies, for all sufficiently large j ,

$$j \ll \log \ell_j \ll j \log(2j), \quad \log \log \ell_j \asymp \log(2j). \quad (3.3)$$

The two-sided comparison will be used when a logarithmic factor has a negative exponent.

Set

$$x_j = (b^{t_j}, b^{r_j}, a_j), \quad \Psi_j = b^{t_j}. \quad (3.4)$$

In particular $\Psi_j = \max_i |(x_j)_i|$. If $g_j = \gcd(b^{r_j}, a_j)$, then

$$H(x_j) = b^{t_j} / g_j, \quad b^{s_j} \leq H(x_j) \leq b^{t_j}. \quad (3.5)$$

Thus the stronger selection of the periods also implies $H(x_{j+1}) > H(x_j)^2$. The argument below, however, uses Ψ_j as a size parameter without identifying it with $H(x_j)$.

4 Uniform intersection with linear subspaces

The next proposition isolates the arithmetic property of the selected sequence that will be used in the counting argument.

Proposition 4.1. Fix $b \geq 2$ and a real algebraic irrational $\xi \in (0, 1)$. Suppose that integer vectors $x_j = (b^{t_j}, b^{r_j}, a_j)$ satisfy

$$0 \leq r_j < t_j, \quad 0 \leq a_j < b^{t_j}, \quad r_j > 0 \implies b \nmid a_j, \quad (4.1)$$

$$|(b^{t_j} - b^{r_j})\xi - a_j| \leq 1, \quad (4.2)$$

$$t_{j+1} > 2t_j, \quad s_j := t_j - r_j \text{ are pairwise distinct.} \quad (4.3)$$

There is a constant $B_{b,\xi}$ such that every proper linear subspace of $\overline{\mathbb{Q}}^3$ contains at most $B_{b,\xi}$ of the vectors x_j . The constant is independent of the subspace and of the sequence.

Proof. Distinct vectors are not proportional. Indeed, if $t' > t$ and $x' = \lambda x$, the first coordinates give $\lambda = b^{t'-t}$, and the second give $r' = r + t' - t > 0$. The third coordinate then gives $b \mid a'$, contradicting (4.1).

Consider a rational plane with primitive integer normal vector $z = (z_1, z_2, z_3) \neq 0$, and put $Z = \max_i |z_i|$. If the plane contains at least two selected vectors, let x_i, x_j , $i < j$, be its first two. Their cross product is a nonzero integer multiple of z . Since all coordinates of x_h have absolute value at most b^{t_h} , we have

$$Z \leq 2b^{t_i+t_j} \leq 2b^{2t_j}. \quad (4.4)$$

If $z_3 = 0$, the equation of the plane is $z_1 b^t + z_2 b^r = 0$. When it has a selected solution, both z_1, z_2 are nonzero and $b^{t-r} = -z_2/z_1$ is fixed. The distinctness of the s_j therefore permits at most one selected vector.

Suppose that $z_3 \neq 0$, and write $b = \prod_{p|b} p^{e_p}$. At a point with $r > 0$, choose a prime $p \mid b$ such that $v_p(a) < e_p$, as allowed by $b \nmid a$. The plane equation implies $b^r \mid z_3 a$, so

$$e_p r \leq v_p(z_3) + v_p(a) \leq v_p(z_3) + e_p - 1.$$

It follows, with $\kappa_b = \min_{p|b} e_p \log p > 0$, that

$$r \leq \frac{\log Z}{\kappa_b} + 1. \quad (4.5)$$

This also holds for $r = 0$. The prime used in this argument is allowed to depend on the point.

Let $A_d > 0$ be the leading coefficient of the primitive minimal polynomial of ξ , and let $\xi = \xi_1, \xi_2, \dots, \xi_d$ be its conjugates. Since $A_d \xi$ is an algebraic integer, $A_d^d N_{\mathbb{Q}(\xi)/\mathbb{Q}}(z_1 + z_3 \xi)$ is a nonzero integer. The other factors in this norm have absolute value at most $Z(1 + |\xi_h|)$. Hence

$$|z_1 + z_3 \xi| \geq c_\xi Z^{-(d-1)}, \quad c_\xi = \left(A_d^d \prod_{h=2}^d (1 + |\xi_h|) \right)^{-1} > 0. \quad (4.6)$$

Write $a = \xi(b^t - b^r) + E$, where $|E| \leq 1$. Substitution into the plane gives

$$(z_1 + z_3 \xi)b^t + (z_2 - z_3 \xi)b^r + z_3 E = 0.$$

Using (4.6) and $b^r \geq 1$, we obtain

$$b^{t-r} \leq c_\xi^{-1}(2 + \xi)Z^d. \quad (4.7)$$

Together, (4.5) and (4.7) imply

$$t \leq \left(\kappa_b^{-1} + \frac{d}{\log b} \right) \log Z + O_{b,\xi}(1).$$

By (4.4), every selected vector in this plane consequently satisfies $t \leq C_{b,\xi} t_j$ for a fixed $C_{b,\xi} \geq 1$. The doubling condition permits only a bounded number of indices after j with this property. There is just one earlier selected vector in the plane, by the choice of i, j . This proves a uniform bound for rational planes and therefore for proper rational subspaces.

Finally, if $T \subsetneq \overline{\mathbb{Q}}^3$ is linear, then $\dim_{\mathbb{Q}}(T \cap \overline{\mathbb{Q}}^3) \leq 2$: otherwise three rationally independent rational vectors in T would have nonzero determinant. If this dimension is 2, the rational-plane bound applies. If it is 1, non-proportionality permits at most one selected vector; if it is 0, there is none, since each x_j is nonzero. Taking $B_{b,\xi} \geq 1$ proves the assertion in every case. $\square$

In particular, $B_{b,\xi}$ is independent of the plane, the chosen sequence, its length N , the approximation parameter ε , and the ratio interval I_l . The exceptional subspace in the application below may change with N and l ; this uniformity permits the same bound to be used for all of them.

Corollary 4.2. *For $b = 2$ and $\xi = \sqrt{2} - 1$, the constant in Proposition 4.1 may be taken to be 4.*

Proof. Use the notation of the proof of Proposition 4.1. Only the case $z_3 \neq 0$ requires consideration. If $r > 0$, then a is odd and the plane equation gives $2^r \mid z_3$. Thus $2^r \leq Z$, also for $r = 0$. The conjugate of ξ is $-\sqrt{2} - 1$, so (4.6) gives $|z_1 + z_3\xi| \geq ((2 + \sqrt{2})Z)^{-1}$. The same plane equation now yields

$$2^{t-r} \leq (2 + \sqrt{2})(\sqrt{2} + 1)Z^2 = (4 + 3\sqrt{2})Z^2 < 16Z^2.$$

Consequently $t < 3 \log_2 Z + 4 \leq 6t_j + 7$, where x_i, x_j are the first two selected vectors in the plane. Since the t_h are integers, $t_{h+1} \geq 2t_h + 1$, and hence $t_{j+3} \geq 8t_j + 7 > 6t_j + 7$. There are no selected vectors of the plane with index at least $j + 3$. At most $x_i, x_j, x_{j+1}, x_{j+2}$ can occur. $\square$

Remark 4.3. The hypotheses of Proposition 4.1 require only the bounded error (4.2). In particular, its constant remains uniform as the approximation exponent tends to zero. The fractional expansions of $\sqrt{2}$ and $\sqrt{2} - 1$ coincide, so Corollary 4.2 applies to the approximations used for $p_{\sqrt{2},2}$.

5 A twisted-height interval theorem

We use the following immediate consequence of [3, Theorem 2.3] in dimension three. Let K be a number field and M_K its set of places. Use absolute values $\|\cdot\|_v$ normalised for the product formula. For every $v \in M_K$, let L_{1v}, L_{2v}, L_{3v} be linearly independent linear forms in $K[X_1, X_2, X_3]$. Assume that they are X_1, X_2, X_3 at all but finitely many places, and that at most R distinct forms occur altogether. Let real weights c_{iv} have finite support in v and satisfy

$$\sum_{i=1}^3 c_{iv} = 0 \quad (v \in M_K), \quad \sum_{v \in M_K} \max_{1 \leq i \leq 3} c_{iv} \leq 1. \quad (5.1)$$

For $Q \geq 1$ and $x \in K^3$, define

$$\mathcal{H}_{\mathcal{L},c,Q}(x) = \prod_{v \in M_K} \max_{1 \leq i \leq 3} (\|L_{iv}(x)\|_v Q^{-c_{iv}}), \quad \mathcal{L} = (L_{iv}).$$

The normalised extension of the forms and weights defines the same height on $\overline{\mathbb{Q}}^3$; only rational vectors will be needed here. If $F_1, \dots, F_R$ is the list of distinct forms, with repetitions added if

needed, put

$$\Delta_{\mathcal{L}} = \prod_{v \in M_K} \|\det(L_{1v}, L_{2v}, L_{3v})\|_v,$$

$$H_{\mathcal{L}} = \prod_{v \in M_K} \max_{1 \leq i_1, i_2, i_3 \leq R} \|\det(F_{i_1}, F_{i_2}, F_{i_3})\|_v.$$

Corollary 5.1 (of Evertse–Ferretti’s interval theorem). *Under these hypotheses, let $0 < \theta \leq 1$ and set*

$$m = \left\lfloor 10^5 2^6 3^{10} \theta^{-2} \log(3\theta^{-1}R) \right\rfloor, \quad \omega = \theta^{-1} \log(3R), \quad C_0 = \max(H_{\mathcal{L}}^{1/R}, 3^{1/\theta}).$$

There is a proper linear subspace $T \subsetneq \overline{\mathbb{Q}}^3$, defined over K , and there are real numbers $Q_1, \dots, Q_m \geq C_0$ such that, for every $Q \geq 1$ and $x \in \overline{\mathbb{Q}}^3 \setminus T$,

$$\mathcal{H}_{\mathcal{L},c,Q}(x) \leq \Delta_{\mathcal{L}}^{1/3} Q^{-\theta} \implies Q \in [1, C_0) \cup \bigcup_{h=1}^m [Q_h, Q_h^{\omega}).$$

Proof. For fixed Q , the existence of an $x \notin T$ satisfying the displayed inequality says exactly that its entire solution set is not contained in T . Apply [3, Theorem 2.3, equations (2.24)–(2.25)]. $\square$

The parameter Q in this corollary is external; it need not equal $H(x)$. The weights are fixed when the theorem is applied. Both the exceptional subspace and the interval endpoints may depend on these weights. The number of intervals is $O_R(\theta^{-2} \log(2/\theta))$.

6 Local inequalities and the height parameter

Return to the sequence (3.4) obtained under (2.3). We keep $\Psi_j = b^{t_j}$ as a size parameter; it is not identified with $H(x_j)$. The parameter used in Corollary 5.1 will be $Q_j = \Psi_j^{1+\delta/3}$, so common divisors of the integer coordinates require no additional partition. Fix a sufficiently large even integer N , and put

$$\varepsilon = a_0/F(\ell_N), \quad k = \lceil 2/\varepsilon \rceil, \quad \delta = \varepsilon - 1/k, \quad \theta = \frac{\delta}{3 + \delta}. \quad (6.1)$$

For large N , one has

$$0 < \varepsilon \leq 1, \quad \varepsilon/2 \leq \delta \leq \varepsilon, \quad \varepsilon/8 \leq \theta \leq \varepsilon/3. \quad (6.2)$$

For every $j \leq N$, Lemmas 2.2 and 2.3 give

$$v_j \geq a_0 \ell_j / F(\ell_j) \geq \varepsilon t_j, \quad |(b^{t_j} - b^{r_j})\xi - a_j| \leq \Psi_j^{-\varepsilon}. \quad (6.3)$$

Partition $[0, 1)$ into the k intervals $I_l = [l/k, (l+1)/k)$, $0 \leq l < k$, and assign a point to I_l when $\rho_j = r_j/t_j \in I_l$. Fix one such interval. For $p \mid b$, put

$$w_p = \frac{e_p \log p}{\log b}, \quad \sum_{p \mid b} w_p = 1, \quad S = \{\infty\} \cup \{p : p \mid b\}.$$

Here $|p|_p = p^{-1}$ and $|\cdot|_\infty$ is the usual real absolute value. Choose an extension of each to $\overline{\mathbb{Q}}$. At the rational places in S , consider the forms and exponents

$$\begin{array}{c|ccc|ccc} & L_{1,p} & L_{2,p} & L_{3,p} & e_{1,p} & e_{2,p} & e_{3,p} \\ \hline \infty & X_1 & X_2 & -\xi X_1 + \xi X_2 + X_3 & 1 & (l+1)/k & -\varepsilon \\ p \mid b & X_1 & X_2 & X_3 & -w_p & -w_p l/k & 0. \end{array} \quad (6.4)$$

Every selected vector assigned to I_l satisfies

$$|L_{i,p}(x_j)|_p \leq \Psi_j^{e_{i,p}} \quad (p \in S, 1 \leq i \leq 3). \quad (6.5)$$

At infinity this follows from the upper endpoint of I_l and (6.3). At $p \mid b$ it follows from

$$|b^{t_j}|_p = \Psi_j^{-w_p}, \quad |b^{r_j}|_p = \Psi_j^{-w_p \rho_j} \leq \Psi_j^{-w_p l/k}, \quad |a_j|_p \leq 1.$$

These are inequalities for integer coordinates without primitive normalisation. Their exponents satisfy

$$\sum_{p \in S} \sum_i e_{i,p} = -\delta, \quad \sum_{p \in S} \max_i e_{i,p} = 1. \quad (6.6)$$

For completeness, we give the conversion to twisted height, following the construction in [2, §5, equations (5.14)–(5.17)]. Choose a finite Galois extension $K/\mathbb{Q}$ containing ξ and its conjugates. The previously fixed extensions $|\cdot|_p$ on $\overline{\mathbb{Q}}$ serve as reference absolute values. For v above a rational place p , choose $\sigma_v \in \text{Gal}(K/\mathbb{Q})$ such that

$$\|z\|_v = |\sigma_v(z)|_p^{d_v} \quad (z \in K), \quad d_v = \frac{[K_v : \mathbb{Q}_p]}{[K : \mathbb{Q}]}, \quad \sum_{v|p} d_v = 1. \quad (6.7)$$

Such a choice exists because the Galois group acts transitively on the places of K above p . Make these choices once, independently of the points and their parameters. At infinity $\mathbb{Q}_\infty = \mathbb{R}$, with the corresponding real or complex local degree. This is the normalisation of [3, §1.1 and §2.1, equations (2.1)–(2.3)]; in particular a complex place has local degree 2.

Lemma 6.1 (Transport of a local form). *For $v \mid p$, every linear form $L \in K[X_1, X_2, X_3]$ and every $x \in \mathbb{Q}^3$ satisfy*

$$\|\sigma_v^{-1}(L)(x)\|_v = |L(x)|_p^{d_v}. \quad (6.8)$$

Here σ_v^{-1} acts on the coefficients of the form.

Proof. The automorphism σ_v fixes each coordinate of x . Thus

$$\sigma_v(\sigma_v^{-1}(L)(x)) = L(\sigma_v(x)) = L(x).$$

Applying (6.7) proves (6.8). □

At places above S set

$$\tilde{L}_{iv} = \sigma_v^{-1}(L_{i,p}), \quad \tilde{e}_{iv} = d_v e_{i,p};$$

at all other places set $\tilde{L}_{iv} = X_i$ and $\tilde{e}_{iv} = 0$. For $v \mid p$ with $p \in S$, Lemma 6.1 and (6.5) give the exact comparison

$$\|\tilde{L}_{iv}(x_j)\|_v = |L_{i,p}(x_j)|_p^{d_v} \leq \Psi_j^{d_v e_{i,p}} = \Psi_j^{\tilde{e}_{iv}}.$$

Together with integrality at places not above S , this proves

$$\|\tilde{L}_{iv}(x_j)\|_v \leq \Psi_j^{\tilde{e}_{iv}} \quad (v \in M_K). \quad (6.9)$$

Remark 6.2. For example, take $K = \mathbb{Q}(\sqrt{2})$, $\xi = \sqrt{2} - 1$, and $\xi' = -\sqrt{2} - 1$. Let $\tau(\sqrt{2}) = -\sqrt{2}$, and let v_+, v_- be the two real places. With the positive embedding as reference,

$$\|z\|_{v_+} = |z|^{1/2}, \quad \|z\|_{v_-} = |\tau(z)|^{1/2}.$$

For $q = b^t - b^r$ and $x = (b^t, b^r, a)$, the forms chosen above give

$$\tilde{L}_{3,v_+}(x) = a - q\xi, \quad \tilde{L}_{3,v_-}(x) = a - q\xi'.$$

Their corresponding local absolute values are both $|a - q\xi|^{1/2}$, since $\tau(a - q\xi') = a - q\xi$. Although $|a - q\xi'|$ can be large in the reference embedding, this is not its absolute value at v_- . The forms used at the two places are different, as permitted in the definition of twisted height.

All local determinants are 1. The forms belong to the list X_1, X_2, X_3 and the d conjugates of $-\xi X_1 + \xi X_2 + X_3$. Consequently $\Delta_{\mathcal{L}} = 1$ and $R = d + 3$ is admissible; the forms, field, and $H_{\mathcal{L}}$ are independent of N, ε, l . The identities (6.6) lift to

$$\sum_{v,i} \tilde{e}_{iv} = -\delta, \quad \sum_v \max_i \tilde{e}_{iv} = 1.$$

Define

$$\bar{e}_v = \frac{1}{3} \sum_i \tilde{e}_{iv}, \quad c_{iv} = \frac{\tilde{e}_{iv} - \bar{e}_v}{1 + \delta/3}, \quad Q_j = \Psi_j^{1+\delta/3}. \quad (6.10)$$

The weights have finite support, sum to zero at each place, and obey

$$\sum_v \max_i c_{iv} = \frac{\sum_v \max_i \tilde{e}_{iv} - \sum_v \bar{e}_v}{1 + \delta/3} = 1.$$

Thus (5.1) holds. By (6.9),

$$\|\tilde{L}_{iv}(x_j)\|_v Q_j^{-c_{iv}} \leq \Psi_j^{\bar{e}_v}.$$

Taking maxima and multiplying over the places gives

$$\mathcal{H}_{\mathcal{L},c,Q_j}(x_j) \leq \Psi_j^{-\delta/3} = Q_j^{-\delta/(3+\delta)} = Q_j^{-\theta}. \quad (6.11)$$

This is precisely the inequality required in Corollary 5.1. Conjugating the forms in the passage to K is essential: the argument uses a small error at the specified real embedding, without asserting the same error for every conjugate.

7 Counting and proof of the main theorem

Lemma 7.1. *Let (Q_j) be a sequence of real numbers greater than 1 such that $Q_{j+1} > Q_j^2$. For $A > 1$ and $\omega > 1$, at most $1 + \lceil \log_2 \omega \rceil$ of the parameters Q_j belong to $[A, A^\omega)$.*

Proof. If q such parameters occur, their logarithms increase by a factor greater than 2^{q-1} from the first to the last. The interval condition gives $2^{q-1} < \omega$ when $q \geq 2$, which implies the stated bound. The case $q \leq 1$ is immediate. $\square$

Proposition 7.2. *Assume (2.3), with F as in (2.1), $u > 0$, and $\gamma \geq 0$. Then, for every sufficiently large even N ,*

$$\frac{N}{2} \ll_{b,\xi} \varepsilon^{-3} (\log(2/\varepsilon))^2 \ll_{b,\xi,C,u,\gamma,k_0,\ell_1} N^{3u} (\log(2N))^{3u+2-3\gamma}, \quad (7.1)$$

where ε is defined by (6.1).

Proof. Use the sequence of Lemma 3.1 and the parameters (6.1). By Proposition 4.1, the exceptional subspace supplied by Corollary 5.1, for any fixed interval I_l , contains at most $B_{b,\xi}$ selected vectors.

The parameter sequence (6.10) satisfies $Q_{j+1} > Q_j^2$. By Lemma 7.1, each of the intervals supplied by Corollary 5.1 contains $O_{b,\xi}(\log(2/\theta))$ selected parameters. The parameters below C_0 can be discarded by restricting to the indices $N/2 \leq j \leq N$. Indeed,

$$\theta^{-1} \ll \varepsilon^{-1} \ll F(\ell_N) \ll (\log \ell_N)^u \ll (N \log(2N))^u, \quad \log C_0 = O_{b,\xi}(1) + O(\theta^{-1}), \quad (7.2)$$

whereas $t_j \geq 2^{j-1}t_1$ and $\log Q_j = (1 + \delta/3)t_j \log b$. Thus $Q_j \geq C_0$ for all such indices when N is large, uniformly over the intervals I_l .

It follows from (6.11) and Corollary 5.1 that each I_l contains at most

$$B_{b,\xi} + O_{b,\xi}(\theta^{-2}(\log(2/\theta))^2)$$

of these selected vectors. Summing over $k = O(\varepsilon^{-1})$ intervals and using (6.2), we obtain

$$\frac{N}{2} \ll_{b,\xi} \varepsilon^{-3} (\log(2/\varepsilon))^2 \ll_{b,\xi,C} F(\ell_N)^3 (\log(2F(\ell_N)))^2. \quad (7.3)$$

Put $y_N = \log \ell_N$. Then $F(\ell_N) = y_N^u (\log y_N)^{-\gamma}$ and $\log(2F(\ell_N)) \ll_{u,\gamma} \log y_N$. Consequently the right side of (7.3) is at most a constant times

$$y_N^{3u} (\log y_N)^{2-3\gamma}.$$

By (3.3), $y_N \ll N \log(2N)$ and $\log y_N \asymp \log(2N)$. Since $3u > 0$, this proves the second inequality in (7.1). The two-sided comparison for $\log y_N$ justifies the conclusion also when $2 - 3\gamma < 0$. $\square$

Proof of Theorem 1.1. If the limsup in (1.1) were finite for some $\eta < 1/3$, choose $\max(\eta, 0) < u < 1/3$. The bound (2.3) would then hold with $\gamma = 0$ and suitable fixed C, k_0 . Proposition 7.2 gives $N \ll N^{3u} (\log(2N))^{3u+2} = o(N)$, a contradiction. $\square$

Proof of Theorem 1.2. If the limsup in (1.2) were finite for some fixed $\gamma > 1$, then (2.3) would hold with $u = 1/3$ and that value of γ . Proposition 7.2 would give

$$N \ll N (\log(2N))^{3-3\gamma} = o(N),$$

again a contradiction. $\square$

8 The role of fixed weights

The factor ε^{-1} from the partition of r/t is relevant to the exponent in (7.1). The following elementary observation specifies the restriction imposed by the coordinatewise estimates used here.

Proposition 8.1. *Fix $\varepsilon > 0$. Consider exponent vectors obtained by uniform coordinatewise majorisation of*

$$(1, \rho, -\varepsilon) \quad \text{at infinity}, \quad (-w_p, -w_p \rho, 0) \quad \text{at } p \mid b,$$

for all $\rho \in [A, B] \subset [0, 1]$. Every such fixed exponent system has total sum at least $-\varepsilon + B - A$. To retain a negative total sum of magnitude at least $\varepsilon/2$, it is necessary that $B - A \leq \varepsilon/2$.

Proof. The second coordinate requires an exponent at least B at infinity and at least $-w_p A$ at $p \mid b$. The other coordinates require the exponents already displayed. Their total is therefore at least

$$1 + B - \varepsilon - \sum_{p \mid b} w_p(1 + A) = -\varepsilon + B - A. \quad \square$$

This proposition concerns the displayed coordinatewise bounds; it does not exclude estimates that exploit additional relations among the points, or a theorem permitting weights to vary. The parameter theorem used here keeps the weights fixed.

At $u = 1/3$ and $\gamma = 0$, (7.1) becomes $N \ll N(\log(2N))^3$, which is compatible with arbitrarily large N . Theorem 1.2 instead uses $\gamma > 1$. At $\gamma = 1$ the count gives only $N \ll N$; neither this borderline case nor the unweighted endpoint is obtained. The proof also extracts no effective upper bound for a length witnessing (1.1) or (1.2). Its conclusion leaves open both polynomial improvements in L and positive entropy.

A Comparison with covering theorems

The same sequence and uniform subspace bound can be combined with two covering theorems. We record the calculation to distinguish the contribution of the subspace bound from that of the interval theorem. These are weaker consequences of the same argument, not additional assumptions in Theorem 1.1.

First, [2, Theorem 5.1] applies directly to the unnormalised system (6.5). Throughout this comparison take $\gamma = 0$, so that $F(x) = (\log x)^u$. In dimension 3, for fixed forms of bounded degree and height, its number of exceptional rational subspaces is $O_{b,\xi}((1 + \delta^{-1})^7)$. Its required size threshold is

$$\Psi > \max(2H_*, 3^{6/\delta}),$$

where H_* is a fixed bound for the heights $H(1, \alpha_1, \alpha_2, \alpha_3)$ of their coefficient vectors $(\alpha_1, \alpha_2, \alpha_3)$. The hypotheses on exponents are $e_{i,\infty} \leq 1$, $e_{i,p} \leq 0$ at finite places, and total sum $-\delta < 0$. They all hold in (6.4). Here there are four distinct forms, each of degree at most d , and each local determinant is 1. The last half of the selected vectors exceeds the threshold by the same comparison as in (7.2). Proposition 4.1 therefore gives $O_{b,\xi}(\varepsilon^{-7})$ points per interval I_l and $O_{b,\xi}(\varepsilon^{-8})$ in total. Under (2.3), this is

$$N \ll (N \log(2N))^{8u},$$

a contradiction for $u < 1/8$. This combines the covering theorem of [2] with Proposition 4.1 of the present paper; the complexity result stated in [2, Theorem 2.1] itself has range $\eta < 1/11$.

Second, [3, Theorem 2.1] applies to the twisted-height system of Section 6. For fixed R , it gives $O_R(\theta^{-3}(\log(2/\theta))^2)$ proper subspaces covering all solutions with $Q \geq C_0$; more precisely, for each such parameter, the corresponding solutions are contained in one member of this fixed finite collection. All hypotheses and the threshold have already been checked. The resulting total is $O_{b,\xi}(\varepsilon^{-4}(\log(2/\varepsilon))^2)$, and hence

$$N \ll N^{4u}(\log(2N))^{4u+2},$$

which is impossible for $u < 1/4$.

For comparison, the interval theorem used in the main proof gives the following counts. Constants in the table depend on the fixed b, ξ .

Input	Total selected-point bound	Range
[2, Theorem 5.1]	$O(\varepsilon^{-8})$	$\eta < 1/8$
[3, Theorem 2.1]	$O(\varepsilon^{-4}(\log(2/\varepsilon))^2)$	$\eta < 1/4$
[3, Theorem 2.3]	$O(\varepsilon^{-3}(\log(2/\varepsilon))^2)$	$\eta < 1/3$

The theorem numbers for [3] refer to the published version.

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